Difference Equation Descriptions: Solving System Responses with Stored Energy

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Key Concepts

1) Difference equations are one of the few descriptions that allow analysis of a system with stored energy at the time an input is applied.

2) Difference equation

$$\sum_{k=0}^{N} a_k y[n-k] = \sum_{k=0}^{M} b_k x[n-k]; \qquad a_0 = 1$$

   a) Input $x[n]$, output $y[n]$

b) Order of difference equation is N

c) Initial conditions $y[-1], y[-2], \ldots, y[-N]$

i) Summarize the entire past history of the system. Nothing more needs to be known about the past to determine current and future output.

ii) If they are zero, the system is at rest.

iii) If they are non zero, they represent the energy stored in the system due to past inputs.

3) Analytical approach for solving difference equations gives insight

into the relationship between the coefficients $\{b_0, \ldots, b_M, a_1, \ldots, a_N\}$ and the system response.

4) The difference equation is linear so the output can be written as a sum of two terms $y[n] = y_s[n] + y_t[n]$.

    a) Steady state response $y_s[n]$: output when the input is applied for all time.

    b) Transient response $y_t[n]$: how the system transitions from its initial condition to steady state.

5) Steady state response $y_s[n]$.

a) Assume the input is a sum of complex exponential signals

$$x[n] = \sum_{i=1}^{P} \alpha_i z_i^n$$

where z_i is a complex number.

b) We only need the output for an input $x[n] = z^n$. The output is $H(z)z^n$ where

$$H(z) = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}}$$

Then by linearity

$$y_s[n] = \sum_{i=1}^{P} \alpha_i H(z_i) z_i^n$$

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c) $H(z)$ is called the system function.

6) Transient response $y_t[n]$ is the system output when the input is 0. This follows from linearity and the fact that $y_s[n]$ is the output for the specified input.

a) The transient response is

$$y_t[n] = \sum_{\ell=1}^{N} c_\ell d_\ell^n$$

b) The d_ℓ are the N roots of the system's characteristic equa-

tion

$$\sum_{k=0}^{N} a_k z^{-k} = 0$$

c) The characteristic equation can also be written in terms of positive powers of z as

$$\sum_{k=0}^{N} a_k z^{N-k} = 0$$

d) The constants c_ℓ are chosen so that $y[n] = y_s[n] + y_t[n]$ satisfies the initial conditions.

e) This form of the solution applies when the roots d_ℓ are distinct.

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7) The solution $y[n]$ is only valid for $n \geq 0$ so the initial conditions $y[-1], y[-2], \ldots, y[-N]$ must be translated forward by computing $y[0], y[1], \ldots, y[N-1]$ before solving for the constants c_ℓ. This is done using the recursive form of the difference equation

$$y[n] = -\sum_{k=1}^{N} a_k y[n-k] + \sum_{k=0}^{M} x[n-k]$$

for $n = 0, 1, \ldots, N-1$ assuming $x[n] = 0$ for $n < 0$.

8) Given $y[0], y[1], \ldots, y[N-1], y_s[n]$ and $d_1, d_2, \ldots d_N$, we obtain c_ℓ by

solving the system of N linear equations

$$y[0] = c_1 d_1^0 + c_2 d_2^0 + \ldots + c_N d_N^0 + y_s[0]$$

$$y[1] = c_1 d_1^1 + c_2 d_2^1 + \ldots + c_N d_N^1 + y_s[1]$$

$$\vdots$$

$$y[N-1] = c_1 d_1^{N-1} + c_2 d_2^{N-1} + \ldots + c_N d_N^{N-1} + y_s[N-1]$$

Video Screenshots

Difference equation descriptions allow systems with stored energy to be analyzed. 1

N^{th} order system $$\sum_{k=0}^{N} a_k y[n-k] = \sum_{k=0}^{M} b_k x[n-k] \; ; \; a_0 = 1$$

Initial conditions: $y[-1], y[-2], \cdots, y[-N]$

- stored energy if nonzero
- summarize the effect of all past activity

given $y[-1], y[-2], \cdots, y[-N]$ and $x[n]$ $n \geq 0$ we can find $y[n]$, $n \geq 0$

$$y[n] = -\sum_{k=1}^{N} a_k y[n-k] + \sum_{k=0}^{M} b_k x[n-k]$$

Analytical solution gives insight into system characteristics

Use linearity - write output as the sum of

1) Steady-state response - as if input is on forever.

2) Transient response - transition from initial condition to steady state

Steady-State Response

Consider inputs expressed as sums of complex exponentials

$$x[n] = c_1 z_1^n + c_2 z_2^n \quad \text{where } z_1, z_2 \text{ are complex numbers}$$

Linearity: consider $x[n] = z^n$ } Do these satisfy diff. eqn?

$$\Rightarrow y_s[n] = H(z)\, z^n$$

$$\sum_{k=0}^{N} a_k y[n-k] = \sum_{k=0}^{M} b_k x[n-k]$$

Note: $x[n-k] = z^{-k} z^n$

$y_s[n-k] = H(z)\, z^{-k}\, z^n$

$$\sum_{k=0}^{N} H(z) z^n a_k z^{-k} = \sum_{k=0}^{M} z^n b_k z^{-k} \Rightarrow$$

$$H(z) = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}}$$

"system function"

Steady-state response example:

$$y_s[n] - \tfrac{1}{3} y_s[n-1] = 2x[n] - x[n-1]$$

Suppose $x[n] = (-1/2)^n \rightarrow z = -1/2$ $\quad H(z) = \dfrac{2 - z^{-1}}{1 - \frac{1}{3} z^{-1}}$

$$H(-1/2) = \frac{2 + 2}{1 + 2/3} = 12/5 \quad \text{so} \quad y_s[n] = \frac{12}{5} (-1/2)^n$$

Transient Response

$$y[n] = y_s[n] + y_t[n]$$

$y_s[n]$ satisfies diff. eqn for $x[n]$, by linearity $y_t[n]$ satisfies diff. eqn. for input = 0

$$\sum_{k=0}^{N} a_k \, y_t[n-k] = 0$$

$$y_t[n] = \sum_{\ell=1}^{N} c_\ell (d_\ell)^n \quad \text{where } d_\ell \text{ are roots of } \sum_{k=0}^{N} a_k z^{-k} = 0$$

characteristic eqn

Check: $\displaystyle\sum_{k=0}^{N} a_k d^{n-k} = d^n \sum_{k=0}^{N} a_k d^{-k} = d^n \cdot 0$

$y_t[n]$ assumes nonrepeated roots

c_ℓ are constants to be determined

Notes-d_k are the roots of the denominator of $H(z) = \dfrac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}}$

$$\sum_{k=0}^{N} a_k z^{-k} = 0 \Rightarrow z^N \sum_{k=0}^{N} a_k z^{-k} = \sum_{k=0}^{N} a_k z^{N-k} = 0$$

polynomial in positive powers of z

Example: transient response

$$y[n] - 1/3\, y[n-1] = 2x[n] - x[n-1]$$

$y_t[n] - \frac{1}{3} y_t[n-1] = 0 \Rightarrow y_t[n] = C d^n$, d is root of: $1 - 1/3\, z^{-1} = 0$

or $z - 1/3 = 0 \Rightarrow z = 1/3$ so $y_t[n] = C(1/3)^n$

Constants in $y_t[n]$ chosen so $y[n] = y_s[n] + y_t[n]$ satisfy initial conditions

$y[n]$ is only valid for $n \geq 0 \Rightarrow$ translate initial conditions

$y[-1], y[-2], \cdots y[-N]$ forward by computing $y[0], y[1], \cdots y[N-1]$ prior to solving for constants

Example (con't): $y[n] - \frac{1}{3}y[n-1] = 2x[n] - x[n-1]$ for $x[n] = (-\frac{1}{2})^n u[n]$

and $y[-1] = -6$.

Solution: $y[n] = \left\{\frac{12}{5}(-\frac{1}{2})^n + c(\frac{1}{3})^n\right\} u[n]$, Now find c

- Translate $y[-1]$ to $y[0]$ using $y[0] = \frac{1}{3}y[-1] + 2x[0] - x[-1]$

$= \frac{1}{3}(-6) + 2 \cdot 1 - 0 = 0$

- Solve for c

$$0 = \frac{12}{5}(-\frac{1}{2})^0 + c(\frac{1}{3})^0 \Rightarrow c = -\frac{12}{5}$$

$$y[n] = \left\{\frac{12}{5}(-\frac{1}{2})^n - \frac{12}{5}(\frac{1}{3})^n\right\} u[n]$$